

STOCHASTIC PHYSICS-INFORMED NEURAL NETWORK WITH MC DROPOUT FOR ESTIMATING CONSOLIDATION PARAMETERS

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ABSTRACT

Reliable estimation of the coefficient of consolidation is essential for settlement prediction, pore-pressure dissipation analysis, and staged construction planning in soft ground engineering. Conventional oedometer-based methods and inverse back-analysis are often costly, time-consuming, and difficult to update when only sparse field measurements are available. Although recent physics-informed neural network approaches have shown promise for geotechnical parameter estimation, limited attention has been given to uncertainty-aware inverse estimation of consolidation parameters from sparse and noisy spatiotemporal observations, particularly in a form suitable for sensor-based monitoring and future digital twin applications. To address this gap, this study develops a stochastic physics-informed neural network framework that embeds Terzaghi's one-dimensional consolidation equation into the training loss and integrates Monte Carlo dropout for uncertainty quantification. The proposed model estimates the coefficient of consolidation directly from only 50 synthetic sensor readings, with added Gaussian noise, to represent field-like monitoring constraints. The inverse PINN recovered the consolidation coefficient with high accuracy, estimating $c_v = 1.2093$ against the true value of $c_v = 1.20$, corresponding to an absolute error of 0.0093 and a relative error of 0.78%. Evaluation on a dense spatiotemporal grid yielded an MSE of 0.0088, an RMSE of 0.0938, and an MAE of 0.0679. The Monte Carlo dropout intervals achieved 94.67% coverage for the nominal 95% confidence bounds, indicating well-calibrated predictive uncertainty. The results demonstrate that uncertainty-aware PINNs can provide reliable inverse parameter estimation under limited-data conditions while identifying regions where additional monitoring may be required. This framework offers a promising step toward GeoAI-assisted geotechnical monitoring and digital twin systems for soft ground engineering.

Keywords: PINN; PDE, coefficient of consolidation (c_v); Monte Carlo dropout, Geo-Ai

1. INTRODUCTION

Globally, urban centres near the coast and major ports are facing increasing geotechnical and engineering stresses. Because of port expansion, corridor improvements for transport modes, industrial land development, reclamation practices, or creeping urban encroachment into the hillside. The above projects frequently encounter soft, compressible soils, in which the consolidation response is a key factor in serviceability, construction sequencing, and long-term settlement risk. Rapid and robust

estimation of the coefficient of consolidation (c_v) is not an academic exercise in such scenarios. But before we can deliver safe and economical infrastructure (C et al., 2026). Yet geotechnical practice, i.e., oedometer tests and back-analysis, relies on a lengthy, heuristic lab process and limited data collection. Again, those processes are difficult to upgrade. A contemporary GeoAI should thus do much more than just curve fitting: it needs to assimilate limited sensor data, remain consistent with governing mechanics, quantify uncertainty, and update efficiently as information arrives (Murad et al., 2026; Singh et al., 2024). While traditional finite element methods (FEM) remain irreplaceable for high-fidelity forward simulation, they are constrained as an end-to-end inference engine for repeated parameter back-analysis across applications (Sun et al., 2025). Although inverse FEM workflows usually require iterative calibration, mesh generation, constitutive model selection, and repeated solver calls. Therefore, this can be difficult when fast operational decision-making is necessary. At the other extreme, typical black-box deep learning models readily interpolate observations but violate boundary conditions, conservation laws, or diffusion physics when extrapolating beyond the training cloud (Slater et al., 2023). Thus, uncertainty quantification (UQ) is an important aspect of this work (Gal & Ghahramani, 2015). Safety, constructability, and lifecycle cost are affected by pore-pressure dissipation, foundation settlement, and embankment performance. Considering a purely deterministic model, especially given the sparsity and noise inherent in measurements, is an insufficient basis for decision-making. The hybrid method of physics-informed neural networks (PINNs) embeds the governing partial differential equation (PDE) directly into the training process, providing a temperate balance between data and prior knowledge, while Monte Carlo (MC) dropout provides a simple and scalable means to approximate epistemic uncertainty for deep models. Thus, this study proposes a mesh-free stochastic inverse PINN for 1D consolidation and examines its ability to infer (c_v) from sparse, volatile observations.

2. METHODOLOGY

The overall methodology consists of four main stages. First, a synthetic one-dimensional consolidation field is generated using Terzaghi's governing equation (Terzaghi, 1943), and sparse noisy sensor observations are sampled to represent field-like monitoring conditions. Second, an inverse physics-informed neural network is developed to learn the excess pore pressure response while treating the coefficient of consolidation (c_v) as a trainable parameter (Bekele, 2021). Third, the model is trained by minimizing both data mismatch and physics-residual losses using collocation points across the domain (Bai et al., 2023). Finally, Monte Carlo dropout is applied during inference to estimate predictive uncertainty and generate confidence bounds for the recovered consolidation response.

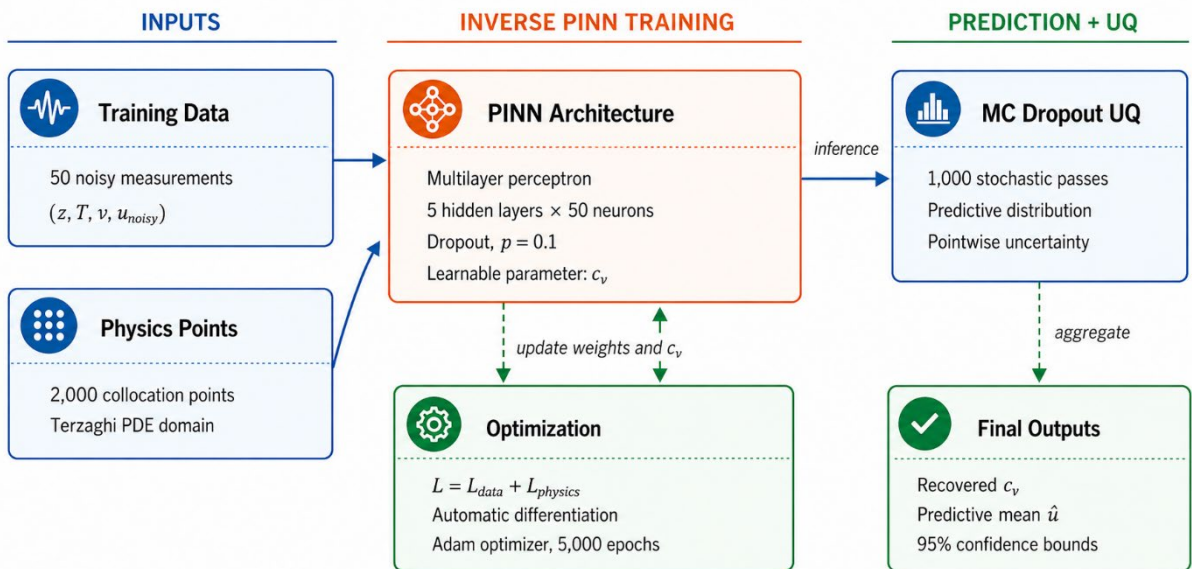


Figure 1: Inverse Physics-Informed Neural Network Framework for Parameter Estimation and Uncertainty Quantification

2.1 Governing Equations

The governing physics is Terzaghi's one-dimensional consolidation equation (Terzaghi, 1943):

$$\frac{\partial u}{\partial t} = c_v \left(\frac{\partial^2 u}{\partial z^2} \right), \quad 0 < z < H, \quad 0 < t \quad (1)$$

where $u(z, t)$ is excess pore pressure, (z) is depth, (t) is the normalised time coordinate reported as (T_v) in the figures, and (c_v) is the coefficient of consolidation to be inferred. In the present study, the depth domain was fixed at $(H = 2.0)$ m, the dimensionless time-factor range was $(0 < T_v < 1)$, and the initial excess pore pressure amplitude was $(u_0 = 1.0)$. The synthetic reference field was generated from the analytical series solution for a double-drainage-type setting with zero excess pore pressure at the drainage boundaries:

$$u(0, t) = 0, \quad u(H, t) = 0, \quad u(z, 0) = u_0 \quad (2)$$

The analytical truth used throughout the study was adopted ($c_v = 1.2$).

2.2 Synthetic data generation under sparse field-like conditions

To simulate real-world instrumentation constraints, the dense analytical field was not used directly for training. 50 spatiotemporal sensor observations were sampled from the analytical solution. Sensor depths followed a uniform density across the soil layer, and time points were drawn from a beta distribution with a small left-hand limit to favour early stages, consistent with earlier statements that early-time consolidation data points appear more informative than later time points. Orthogonal Gaussian noise with a standard deviation of 0.05 was introduced to mimic field conditions (Cao et al., 2018). We obtained a dataset that presents a very difficult inverse problem, with sparse, irregular, and noisy observations. The realised perturbations shown in Figure 2 had a mean of +0.003 and a standard deviation of 0.047, which closely matched the specified noise level.

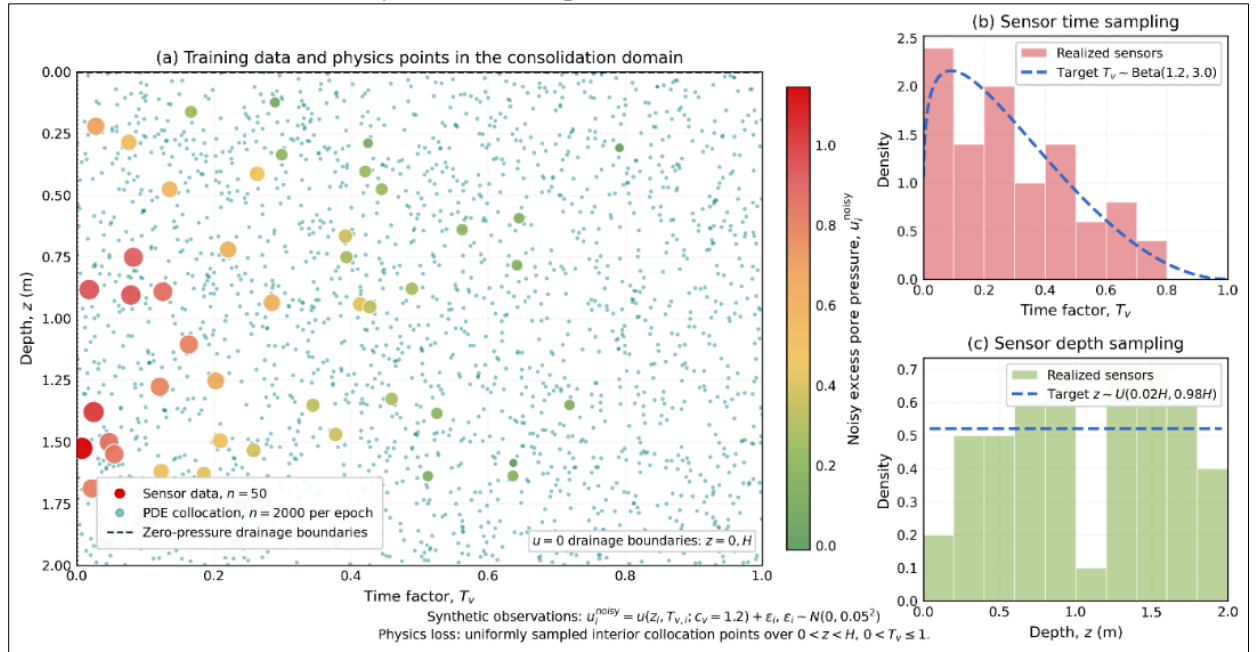


Figure 2: Sparse spatiotemporal sensor and Collocation Sampling in the Consolidation Domain.

2.3 Uncertainty-aware PINN architecture

In this case, the inverse model is implemented in PyTorch in the form of a fully connected multilayer perceptron with five hidden layers (50 neurons each) and hyperbolic tangent activations, along with dropout ($p = 0.1$) (Wu et al., 2023). The inputs (z, t) are scaled to the range of $[-1, 1]$, and $u(z, t)$ is excess pore pressure. Then, the coefficient of consolidation is trained as a positive, regularised parameter (initial value set to 0.5), so that both the state field and the physics laws are learned

simultaneously from incomplete data. Monte Carlo dropout is also applied during inference, performing 1,000 stochastic forward passes to build an empirical predictive distribution (Zhu et al., 2019). The mean gives the final estimate and standard deviation (σ_u) defines approximate 95% confidence bounds. This turns a deterministic PINN into a stochastic inference framework that can identify both regions with strong constraints and those where caution is warranted.

$$\hat{u}(z, t) \pm 1.96\sigma_u(z, t) \quad (3)$$

2.4 Inverse physics-informed loss and optimisation

The network was trained by minimising a composite objective made of a data-misfit term and a physics-residual term:

$$L = L_{data} + L_{phys} \quad (4)$$

The data component was the mean squared error between predictions and the 50 noisy sensor values:

$$L_{data} = \left(\frac{1}{N_s}\right) \sum_{i=1}^{N_s} (u_i - \hat{u}_i)^2 \quad (5)$$

The physics term enforced consistency with the governing PDE through the residual

$$r(z, t) = \frac{\partial u}{\partial t} - c_v \frac{\partial u^2}{\partial z^2} \quad (6)$$

evaluated at 2,000 randomly sampled interior collocation points per epoch:

$$L_{phys} = \left(\frac{1}{N_r}\right) \sum_{j=1}^{N_r} r(Z_j, t_j)^2 \quad (7)$$

Automatic differentiation provided exact first- and second-order derivatives via the computational graph, eliminating the need for manual PDE discretisation. Optimisation was performed with Adam for 5,000 epochs, starting from a learning rate of (10^{-3}) with stepwise decay. To contextualise the value of the physics prior, a degree-8 polynomial regression model trained only on the noisy sensor data was also evaluated as a non-physics baseline (Rao et al., 2023).

3. RESULTS AND DISCUSSION

3.1 Convergence and inverse parameter discovery

The inverse PINN estimated $c_v = 1.2093$, with a small absolute error of 0.0093 and a relative error of 0.78% compared to the true value (1.2). Training remained stable over the 5,000 epochs we trained for, with data loss consistently decreasing from 0.299 to 0.020, while physics loss stayed near that value at ≤ 0.010 . The PDE residual initially dropped rapidly, before levelling out at around 10^{-2} demonstrating that the model learned from noisy data, returning the underlying physics. The estimated c_v rose non-linearly from 0.5 and approached the true value by the last time point. With just 50 noisy samples, the model still predicted accurately thanks to the helpful smearing provided by physics-based regularisation, an example of it working well. The model focused on the full domain yielded (MSE = 0.0088, RMSE = 0.0938 and MAE = 0.0679). The training sensor error (MSE = 0.0165) suggests the model generalizes well and balanced data fitting with physical fidelity without overfitting.

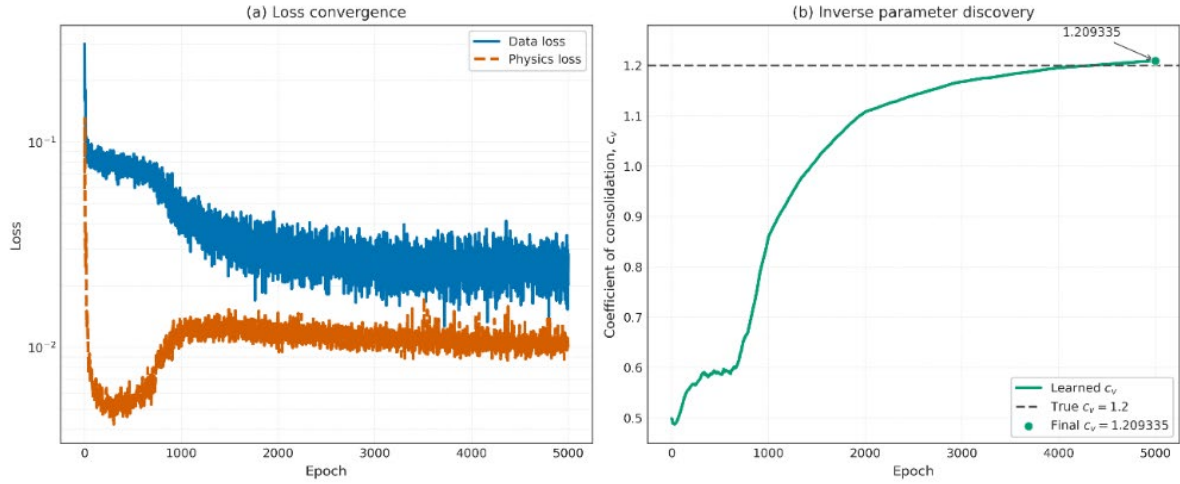


Figure 3: Training history for data loss, physics loss, and learned c_v , showing stable convergence toward the true value of 1.2.

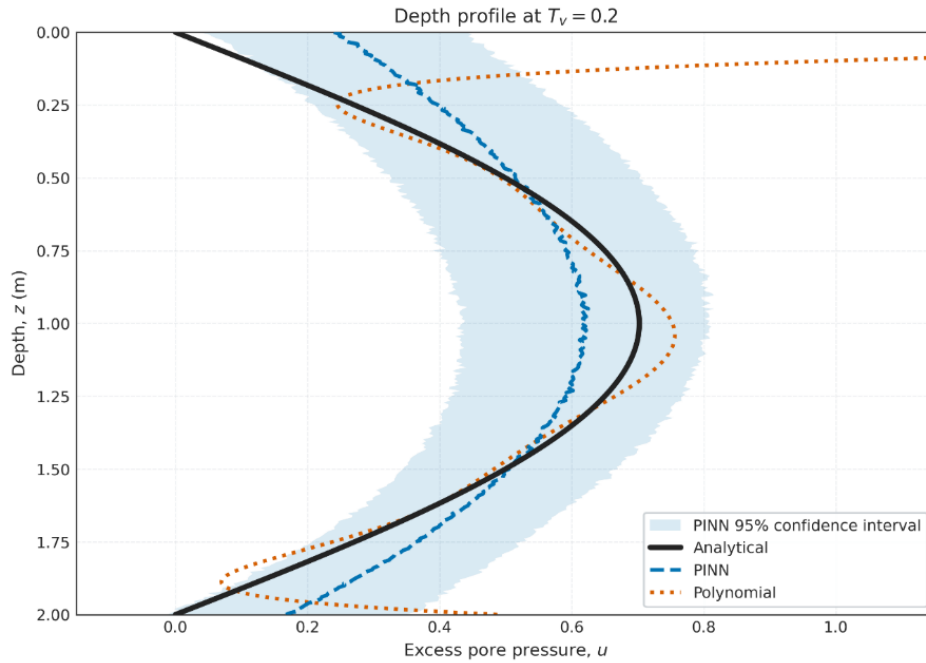


Figure 4: Depth profile at $T_v=0.2$ illustrates that the PINN aligns 95% with the physics-based reference, while the polynomial baseline diverges in poorly constrained areas.

3.2 Uncertainty structure and confidence calibration

The average prediction uncertainty (standard deviation with MC dropout) was 0.085. The maximum uncertainty of 0.118 occurred at depth $z = 0.667$ m, which was the initial time ($T_v = 0$). Even with the limited data available, the model's uncertainty estimates are fairly trustworthy, as approximately 94.67% of true values fell within the 95% confidence intervals. The uncertainty is not isotropic: it is lowest in the interior zone, where data and the PDE are informative for the model, while it grows near boundaries and in early-time zones, where less information is available. This pattern can also assist engineers in determining whether they need to install additional sensors (piezometers) at a few specific locations.

3.3 Error localisation and engineering interpretation

The stochastic PINN is useful in practice as it yields low predictive error in the main region of interest (i.e., where most engineering decisions are performed). Yet it also very much reveals its own limitations in fast-changing areas. It is important to understand the distinction between uncertainty and error: uncertainty provides a conservative overall indication of risk, while error indicates a mistake at a local scale. The two work in tandem to help engineers make better choices about where to locate sensors and how to observe the system. Moreover, it is a mesh-free method that runs quickly on a CPU and thus provides an efficient implementation for real-time applications critical to large infrastructure systems.

3.4 Scope and limitations

We demonstrate a proof of concept that the stochastic PINN can indeed faithfully recover consolidation parameters from sparse, noisy data whilst adhering to the governing physics. The results reaffirm that the core method is validated, with simplified assumptions limiting its field applications. This study is limited to a simplified one-dimensional consolidation problem and validated with synthetic data. The methodology assumes homogeneous soil conditions and ideal boundary conditions. That may not describe field complexity. For uncertainty quantification, we use approximate Bayesian inference. Future research will be conducted with complete probabilistic methods. Additionally, validation will be with real field data.

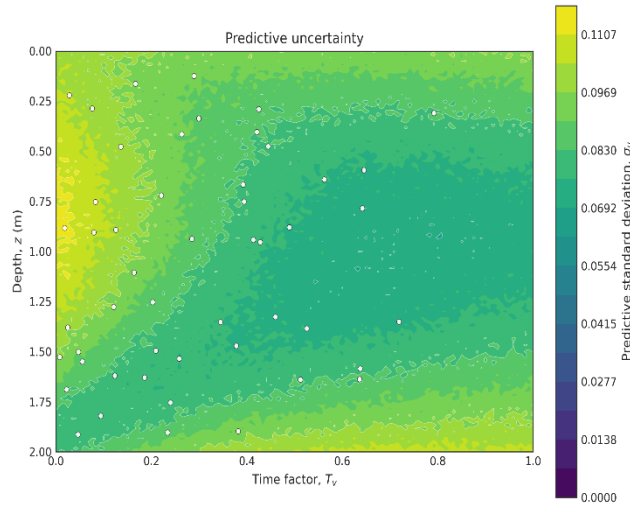


Figure 5: MC-dropout uncertainty map over depth and time, highlighting regions where more data is needed to improve confidence.

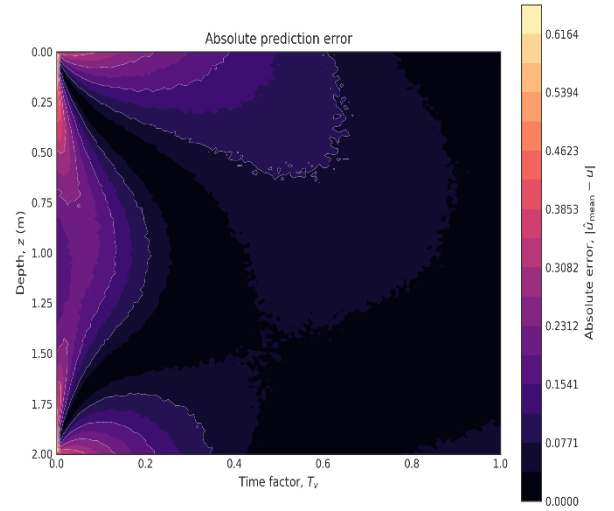


Figure 6: Absolute error of the PINN is greater near early-time and boundary regions, while lower across most of the interior.

4. CONCLUSION

- Demonstrated that an uncertainty-aware Physics-Informed Neural Network (PINN) can successfully estimate consolidation parameters using only a small number of noisy sensor measurements.
- Ensured predictions remained consistent with the underlying governing physical laws despite limited data availability.
- Accurately recovered the key consolidation parameter, obtaining a value very close to the true parameter.
- Achieved strong predictive performance across most of the modeled domain.

- Quantified prediction uncertainty, identifying regions where model predictions are more or less reliable.
- Showed that physics-informed machine learning can serve as a practical tool for geotechnical parameter interpretation when extensive field or laboratory testing is unavailable.
- Highlighted the potential of the framework for real-time monitoring systems and digital twin applications in locations such as Chittagong, where rapid and data-efficient decision-making is essential.
- Established a foundation for future research involving more complex soil conditions and validation with real-world field data.
- Provided an important proof-of-concept and first demonstration of this uncertainty-aware PINN approach within the geotechnical engineering paradigm.

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